

NEET Biology

Detailed NEET-UG Syllabus

Exam Coverage	NEET-UG aligned topic map for complete subject planning
Format	Unit-wise syllabus with detailed bullet-point subtopic breakdown
Use Case	Course brochure, batch syllabus design, chapter planning and academic structuring

Structured Syllabus

Unit 1: Diversity in Living World

- The living world, biological classification, taxonomy and systematics.
- Five kingdom classification, viruses, viroids, lichens and detailed plant/animal kingdom overview.

Unit 2: Structural Organisation in Animals and Plants

- Morphology and anatomy of flowering plants, tissues and tissue systems.
- Structural organization in animals including cockroach, frog and animal tissues.

Unit 3: Cell Structure and Function

- Cell theory, prokaryotic and eukaryotic cells, cell organelles and biomolecules.
- Cell cycle, mitosis, meiosis and transport across cell membrane.

Unit 4: Plant Physiology

- Transport in plants, mineral nutrition, photosynthesis, respiration and plant growth regulators.
- Transpiration, translocation, nitrogen metabolism and growth/development in plants.

Unit 5: Human Physiology

- Digestion and absorption, breathing and exchange of gases, body fluids and circulation.
- Excretory products and elimination, locomotion and movement, neural control and chemical coordination.

Unit 6: Reproduction

- Reproduction in organisms, sexual reproduction in flowering plants.
- Human reproduction, reproductive health, menstrual cycle, fertilization, pregnancy and contraceptive methods.

Unit 7: Genetics and Evolution

- Principles of inheritance and variation, Mendelian genetics, linkage and mutation.
- Molecular basis of inheritance, DNA replication, transcription, translation and genetic code.
- Evolution, origin of life, evidences, Hardy-Weinberg principle and human evolution.

Unit 8: Biology and Human Welfare

- Human health and disease, immunity, common pathogens and vaccines.
- Strategies for enhancement in food production and microbes in human welfare.

Unit 9: Biotechnology and Its Applications

- Principles and processes of biotechnology, recombinant DNA technology, cloning vectors and PCR.
- Applications in medicine, agriculture and industry; transgenic organisms and ethical issues.

Unit 10: Ecology and Environment

- Organisms and populations, ecosystem structure and function, productivity and nutrient cycling.
- Biodiversity, conservation, environmental issues, pollution, climate change and wildlife protection.